PRECISION IMU CARRIER

ARK SCH16T

Precision 6-DoF MEMS IMU carrier based on the Murata SCH16T-K01

The USA-built ARK SCH16T is a compact carrier for the Murata SCH16T-K01 6-DoF MEMS IMU. It provides a triaxial gyroscope (±300 °/s) and triaxial accelerometer (±8 g) over SPI, with onboard 3.3 V regulation from a 5 V supply.

The 11-pin JST-GH connector matches the ARK PAB SPI pinout for direct use with ARK flight controllers and carriers.



KEY FEATURES

- Murata SCH16T-K01 6-DoF MEMS IMU
- Gyro: ±300 °/s, 0.3 °/h bias instability (typ.)
- Accel: ±8 g (±80 m/s²), ±26 g dynamic range
- 20-bit output data
- SafeSPI v2.0 interface with data-ready and sync
- Onboard 3.3 V regulation from 5 V input
- 11-pin JST-GH matching ARK PAB SPI pinout
- Includes cable

SPECIFICATIONS

ELECTRICAL	
Supply voltage	5 V
Supply current	~46 mA average
IMU supply	3.3 V (onboard regulator)
Interface	SPI

ENVIRONMENTAL	
IMU operating temperature	−40 °C to +110 °C
IMU storage temperature	−50 °C to +150 °C (24 h max)
IMU mechanical shock	3000 g, 0.5 ms, any axis (AEC-Q100)
IMU drop	1.2 m onto concrete (AEC-Q100)

MECHANICAL	
Dimensions	22.5 × 25.0 × 5.9 mm
Weight	2.5 g

AT A GLANCE	
IMU	SCH16T-K01
Gyro	±300 °/s
Accel	±8 g
Interface	SPI
Supply	5 V
Dimensions	22.5 × 25 × 5.9 mm
Weight	2.5 g
Hardware rev	Rev 1

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-sch16t/

CAD github.com/ARK-Electronics/ark-hardware

CONNECTOR PINOUT

PIN	SIGNAL	NOTES
1	VDD_5V	5 V supply
2	SPI_SCK	
3	SPI_MISO	
4	SPI_MOSI	
5	SPI_nCS	
6	NC	
7	SPI_nSYNC	
8	SPI_DRDY	Data ready
9	NC	
10	SPI_nRESET	Reset, 10 kΩ pull-up to 3.3 V
11	GND	

Connector: 11-pin JST-GH (SM11B-GHS-TB). Matches the ARK PAB SPI pinout (nCS2 / DRDY2 unused). Signals are 3.3 V.

NOTES

- IMU environmental ratings are the SCH16T-K01 absolute maximum ratings. No vibration survival limit is specified.
- Includes 11-pin JST-GH cable.

ARK-DS-ARK-SCH16T | Version 1.0, 2026-10-29 | Compiled from ARK Electronics published documentation. IMU ranges, performance, and environmental ratings per the Murata SCH16T-K01 data sheet (Doc. No. 11624 Rev. 6). Specifications are subject to change without notice. © 2026 ARK Electronics LLC.